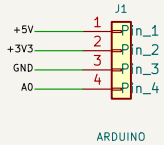
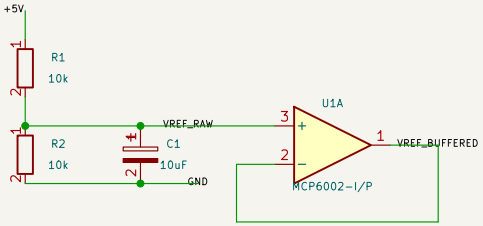
ED4 A ARDUINO INVESTIGATION BOARD

Buffered reference, source control and signal amplifier

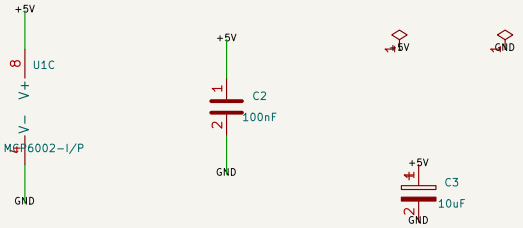
ARDUINO INTERFACE



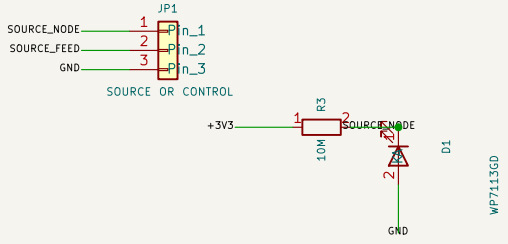
MIDPOINT REFERENCE AND BUFFER

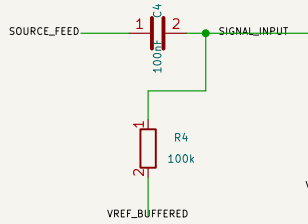


OP AMP POWER

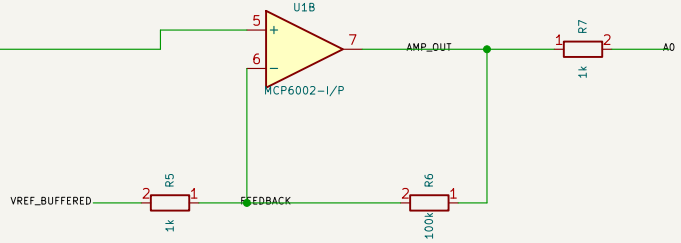


LED ENTROPY SOURCE AND AC COUPLING



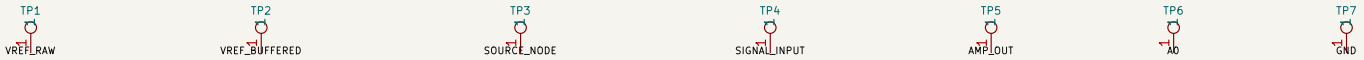


SIGNAL GAIN AND ADC OUTPUT



JP1: 1 to 2 source; 2 to 3 grounded AC control.

BRING-UP TEST POINTS



Nominal U1B gain about VREF: 1 + 100 kOhm / 1 kOhm = 101.

D1 cathode is at SOURCE_NODE. Its anode is at GND.

J1: 1 = 5 V, 2 = 3.3 V, 3 = GND, 4 = A0.

Use a 5 V ADC interface. Arduino AREF stays disconnected.

Arduino AREF remains disconnected
Verify component values and staged measurements before assembly
Connected circuit for the ED4 A Arduino PCB

Trishan Biswas

Sheet: /
File: ED4_A_Arduino.kicad_sch

Title: ED4 A Arduino investigation board

Size: A4	Date: 2026-11-05	**Rev: B**
KiCad E.D.A. 10.0.1		Id: 1/1